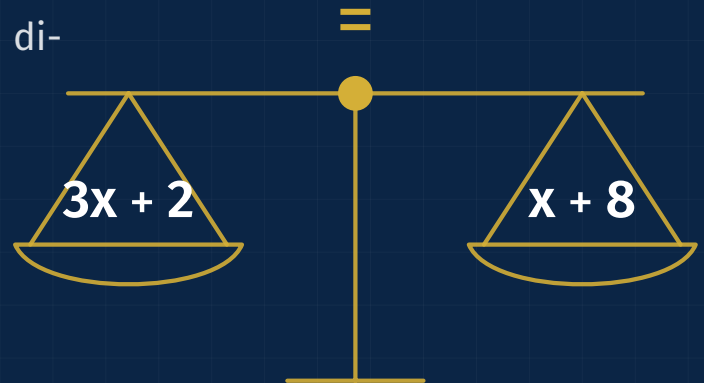


DR ESLAM AHMED
eliteigcse.com

CHAPTER 2 • ALGEBRA • TOPIC 7

Rearranging Formulas

The same two stages as solving an equation — clean, then isolate. When the subject appears twice: collect, factorise, divide.

**6**

VARIANTS

16

PAST-PAPER PARTS

41

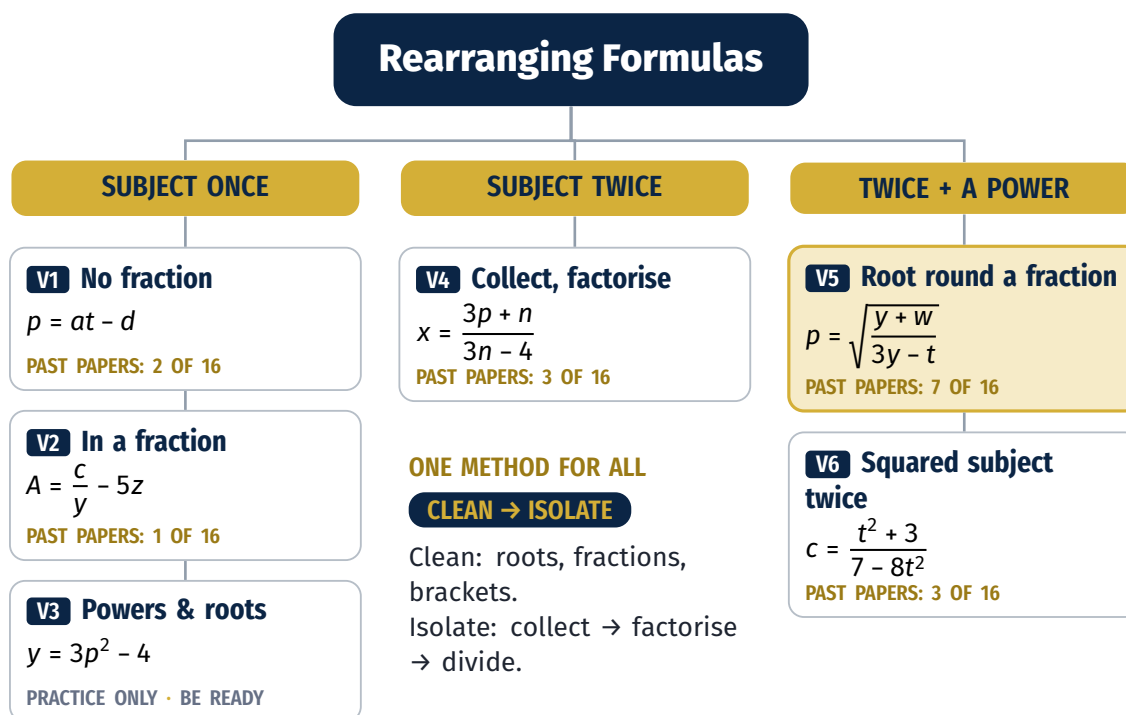
PRACTICE QUESTIONS

YOU WILL MASTER**V1** Subject once**V2** Subject in a fraction**V3** Powers & roots**V4** Subject twice**V5** Root round a fraction**V6** Squared subject twice

TOPIC MAP · EVERY VARIANT ON ONE PAGE

Rearranging Formulas at a glance



Your notes


★ How often does each variant come up?

16 past-paper questions (Jan 2020 → Nov 2025), one rearranging part each; 41 practice parts — grouped by type.

	PAST PAPERS (OUT OF 16)	PRACTICE PARTS
V1 Subject once	2	8
V2 Subject in a fraction	1	6
V3 Powers & roots	0	12
V4 Subject twice: factorise	3	10
V5 Root round a fraction	7	3
V6 Squared subject twice	3	2

WHAT THIS TELLS YOU 13 of 16 past-paper parts have the subject **twice** (V4–V6), 3–4 marks on Paper 2. The skill that wins them: collect, factorise, divide.

BEFORE YOU START

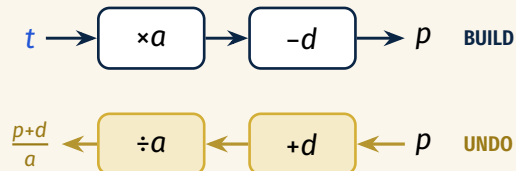
The big idea



Rearranging = solving, with letters.

“Make t the subject” means: get t **alone** on one side. Do exactly what you do to solve an equation — the answer just has letters in it.

Undo in **reverse order**, the opposite operation each time.


Dr Eslam's two stages
STAGE 1
CLEAN

- Root round everything? Square (or cube) both sides
- Fraction? $\times$ the bottom (in a bracket)
- Expand so the subject is free

STAGE 2
ISOLATE

- **Collect** subject terms on one side
- **Factorise** the subject out
- **Divide** by the bracket — then $\pm\sqrt{\quad}$ if squared

DR ESLAM'S RULE · QUADRATIC IN THE SUBJECT

If the subject appears as x^2 **and** x (e.g. $y = x^2 - 4x + 1$), you cannot factorise it out.

Complete the square first (Topic 5): $y = (x - 2)^2 - 3$ — now x appears once:
 $x = 2 \pm \sqrt{y + 3}$. You will meet this again with inverse functions.


CALCULATOR FIRST, THEN CHECK

Check a rearrangement with numbers: pick values (e.g. $a = 2$, $d = 3$, $t = 5$), work out p from the question. Put p , a , d into your answer — you should get $t = 5$ back.

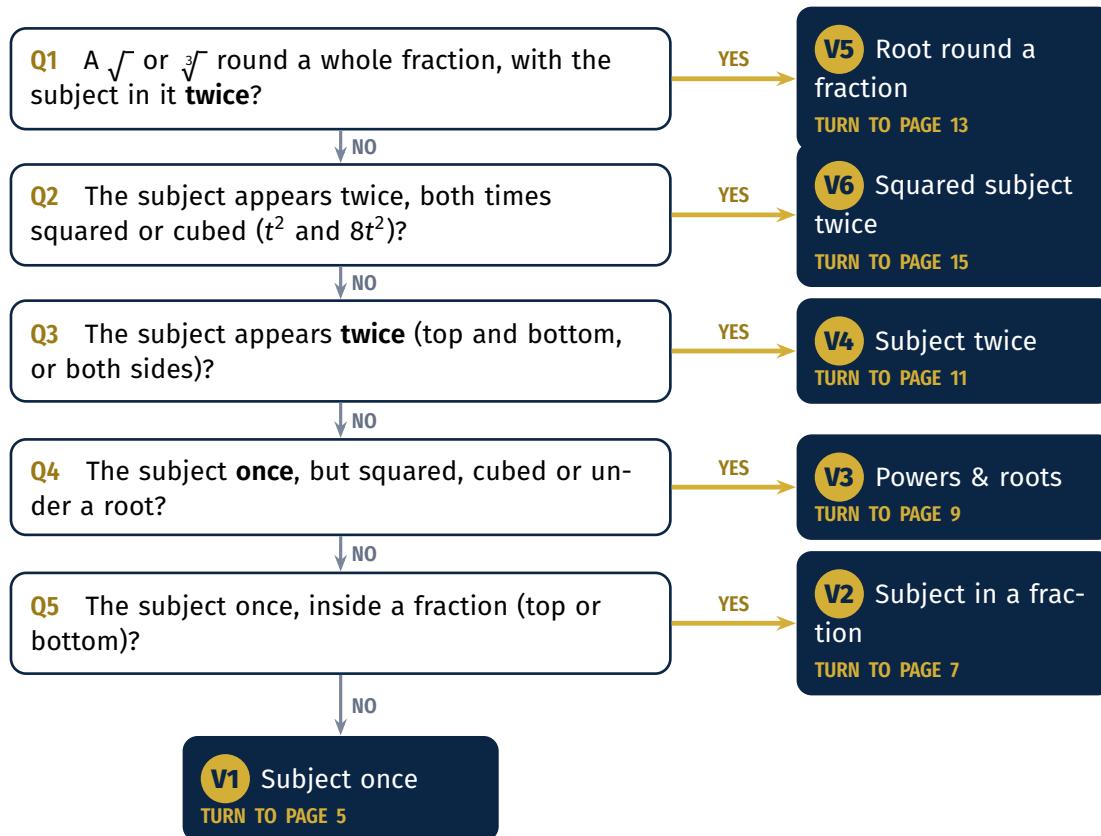
COUNT / RECOGNISE, THEN CHOOSE

Which variant am I?



Your notes

Find the subject. Count how many times it appears, then walk down the ladder.

**COLOUR CODE IN THE WORKING**

Blue = letter terms $4x$, $-3y$

Orange = plain numbers 17 , -13

STEP 1 **STEP 2** **STEP 3** **STEP 4** = which step of "How do I do it?" each line is.

ON EVERY VARIANT SPREAD

- | | | | |
|---|-----------------|---|-----------------------|
|  | What is it? |  | When you see → you do |
|  | How do I do it? |  | Watch out |
|  | Worked example |  | Exam tip |
|  | Check yourself |  | Calculator check |

V1**VARIANT 1****Subject Once**

The subject appears once, no fraction:
undo in reverse order.

SEEN IN PAST PAPERS

2 / 16

PRACTICE QUESTIONS: 8


Your notes

Came up in: Jan 2020 P1HR Q3(a) · May 2021 P1H Q5(b)

? What is it?

In plain words

The subject is in one term. Move everything else **away from it**: first the added/subtracted terms, then divide by what multiplies it.

It is a two-step linear equation — with letters.

IT LOOKS LIKE

$$p = at - d$$

$$d = g + 2ac$$

$$v = u + at$$

$$p = at - d$$

↓ +d BOTH SIDES

$$p + d = at$$

↓ ÷a BOTH SIDES

$$t = \frac{p + d}{a}$$

How do I do it?

1 Find the subject

Circle it. Everything else must move.

2 Add / subtract

Move the terms that are added or subtracted (change sign).

3 Divide

Divide by what multiplies the subject — every term on the other side.

4 Write it

Subject on the left: $t = \dots$

UNDO IN REVERSE ORDER

Last thing done is the first undone.

V1 Subject Once

VARIANT 1 · SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

“Make t the subject of the formula” → **Get t alone: undo in reverse**

“Rearrange ... to make x the subject”
(circle) → **Do it on the side, then circle**

The subject times two letters:
 $d = g + 2ac$ → **Divide by the whole $2c$**

WORKED EXAMPLE **PAST PAPER**

May 2021 · Paper 1H · Q5(b) · 2 marks

Make t the subject of the formula $p = at - d$

STEP 1 $p = at - d$ ◀ find the subject

STEP 2 $p + d = at$ ◀ add d to both sides

STEP 3 $\frac{p+d}{a} = t$ ◀ divide by a

ANSWER $t = \frac{p+d}{a}$ ✓ $a = 2, d = 3, t = 5: p = 7, \frac{7+3}{2} = 5$

WATCH OUT

$$\times t = \frac{p}{a} + d$$

$$\checkmark t = \frac{p+d}{a}$$

Move d first (it was done last), then divide the **whole** side by a .

EXAM TIP

Check with numbers: choose values, find p , put it in your answer — you must get your t back.

Check yourself

calculator first · answers on page 17

1.1

Jan 2020 P1HR

Make a the subject of
 $d = g + 2ac$

Answer:

1.2

Practice · Easy

Make a the subject of
 $M = ac - bd$

Answer:

1.3

Practice · Easy

Make t the subject of
 $v = u + at$

Answer:

V2

VARIANT 2

Subject in a Fraction

The subject is on top or on the bottom of a fraction: clean the fraction first.

SEEN IN PAST PAPERS

1 / 16

PRACTICE QUESTIONS: 6



Your notes

Came up in: Jan 2021 P1HR Q6(a)

? What is it?

In plain words

On top: get the fraction alone, then multiply by its bottom.

On the bottom ($a = \frac{b}{c} + 5$): get the fraction alone, multiply by the subject to bring it up, then divide.

IT LOOKS LIKE

$$A = \frac{c}{y} - 5z$$

$$C = \frac{5(F - 32)}{9}$$

$$a = \frac{b}{c} + 5$$

$$A + 5z = \frac{c}{y}$$

× y BOTH SIDES

$$c = y(A + 5z)$$

The bottom y jumps up and multiplies the **whole** other side.

How do I do it?

1 Fraction alone

Move the other terms away from the fraction.

2 × the bottom

Multiply both sides by the bottom — bracket the other side.

3 Isolate

Finish with add/subtract and divide. Subject on the bottom? It is now on top.

× **THE BOTTOM, BRACKET THE OTHER SIDE** As in Linear Equations V4.

V2 Subject in a Fraction

VARIANT 2 · SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

The subject over a letter: $A = \frac{c}{y} - 5z$ → **Fraction alone, then $\times y$**

The subject in a bracket on top:
 $C = \frac{5(F-32)}{9}$ → **$\times 9, \div 5$, then $+32$**

The subject on the bottom: $a = \frac{b}{c} + 5$ → **Fraction alone, $\times c$, then divide**

WORKED EXAMPLE **PAST PAPER**

January 2021 · Paper 1HR · Q6(a) · 2 marks

Make c the subject of $A = \frac{c}{y} - 5z$

STEP 1 $A + 5z = \frac{c}{y}$ ◀ fraction alone: add $5z$

STEP 2 $y(A + 5z) = c$ ◀ multiply both sides by y

ANSWER $c = y(A + 5z)$ ✓ $y = 2, z = 1, c = 12: A = 1, 2(1 + 5) = 12$

WATCH OUT

✗ $c = yA + 5z$

✓ $c = y(A + 5z)$

The y multiplies the whole side — keep the bracket.

EXAM TIP

$c = yA + 5yz$ is also correct (expanded) — but the bracket form is safer and shorter.

Check yourself

calculator first · answers on page 17

2.1 Practice · EasyMake t the subject of

$$y = \frac{t}{3} - 2a$$

Answer:

2.2 Practice · EasyMake F the subject of

$$C = \frac{5(F - 32)}{9}$$

Answer:

2.3 Practice · MediumMake c the subject of

$$a = \frac{b}{c} + 5$$

Answer:

V3

VARIANT 3

Powers & Roots

The subject is squared, cubed or under a root: undo it with its opposite.

SEEN IN PAST PAPERS

0 / 16

NOT ASKED YET • BE READY
PRACTICE QUESTIONS: 12


Your notes
? What is it?**In plain words**

A power is undone by a root, a root by a power. **Square** undoes $\sqrt{}$; $\sqrt{}$ undoes a square — with $\pm$.

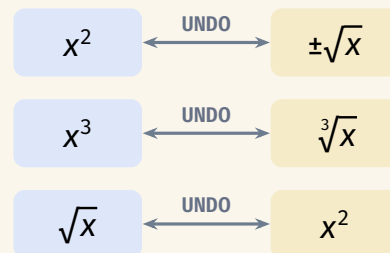
A root round a whole side: **square both sides first**. A squared subject: get t^2 alone, then $\pm\sqrt{}$ **last**.

IT LOOKS LIKE

$$y = 3p^2 - 4$$

$$s = \frac{1}{2}at^2$$

$$m = \sqrt{\frac{k+1}{4}}$$

**How do I do it?****1 Root round all?**

Square (or cube) both sides first.

2 Isolate the powerGet t^2 (or t^3) alone: add/subtract, multiply/divide.**3 Root last** $\pm\sqrt{}$ for a square, $\sqrt[3]{}$ for a cube.

± WHEN YOU SQUARE-ROOT Unless the question says the subject is positive.

V3 Powers & Roots

VARIANT 3 · SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

The subject squared: $y = 3p^2 - 4$ → **Isolate p^2 , then $\pm\sqrt{\quad}$**

A root round the whole side: $m = \sqrt{\frac{k+1}{4}}$ → **Square both sides first**

" $y = \sqrt{w^3}$, make w the subject" → **Square, then cube-root: $w = \sqrt[3]{y^2}$**

**WORKED EXAMPLE****PRACTICE**

Rearranging Formulas · Medium Q4 · 3 marks

Make p the subject of the formula $y = 3p^2 - 4$

STEP 2 $y + 4 = 3p^2$ ◀ add 4: the p^2 term alone

STEP 2 $\frac{y+4}{3} = p^2$ ◀ divide by 3

STEP 3 $p = \pm\sqrt{\frac{y+4}{3}}$ ◀ square root last, $\pm$

ANSWER $p = \pm\sqrt{\frac{y+4}{3}}$

✓ $p = 2$: $y = 8$, $\sqrt{\frac{12}{3}} = 2$

**WATCH OUT**

✗ $p = \sqrt{y} + \frac{4}{3}$

✓ $p = \pm\sqrt{\frac{y+4}{3}}$

Root only at the end, and of the **whole** side — never term by term.

**EXAM TIP**

The $\pm$ is usually needed for full marks unless the question says the subject is positive.

**Check yourself**

calculator first · answers on page 17

3.1

Practice · Easy

Make t the subject of
 $c = t^3 - 8v$

Answer:

3.2

Practice · Medium

Make t the subject of
 $s = \frac{1}{2}at^2$

Answer:

3.3

Practice · Medium

Make k the subject of
 $m = \sqrt{\frac{k+1}{4}}$

Answer:

V4

VARIANT 4

Subject Twice: Collect, Factorise, Divide

The subject is on top and bottom (or both sides): collect it, factorise it out.

SEEN IN PAST PAPERS

3 / 16

PRACTICE QUESTIONS: 10

 Your notes

Came up in: Nov 2021 P1H Q17(b) • Jun 2022 P2H Q14(b) • Nov 2023 P1H Q15

? What is it?

In plain words

The subject appears in **two terms**. You cannot divide it free yet. Clean the fraction ($\times$ the bottom), expand, then **collect** every subject term on one side, **factorise** the subject out, and **divide** by the bracket.

IT LOOKS LIKE

$$x = \frac{3p + n}{3n - 4}$$

$$w = \frac{2f + 3}{8 - f}$$

$$3nx - 4x = 3p + n$$

↓ COLLECT

$$3nx - n = 3p + 4x$$

↓ FACTORISE

$$n(3x - 1) = 3p + 4x$$

↓ DIVIDE

$$n = \frac{3p + 4x}{3x - 1}$$

How do I do it?

1 Clean

$\times$ the bottom (bracket!), then expand every bracket.

2 Collect

All subject terms on one side, everything else on the other.

3 Factorise

Take the subject out as a common factor: $n(3x - 1)$.

4 Divide

Divide by the bracket. Tidy signs if you like.

COLLECT → FACTORISE → DIVIDE Dr Eslam's Isolate stage.

V4 Subject Twice: Collect, Factorise, Divide

VARIANT 4 · SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DOThe subject on top **and** bottom:

$$x = \frac{3p+n}{3n-4}$$

→ × bottom, collect, factorise

The subject on both sides:

$$g - 3m = am + 5$$

→ Collect m terms, factorise $m(a + 3)$ **WORKED EXAMPLE****PAST PAPER**

November 2023 · Paper 1H · Q15 · 3 marks

Make n the subject of the formula $x = \frac{3p+n}{3n-4}$

STEP 1 $x(3n - 4) = 3p + n \Rightarrow 3nx - 4x = 3p + n$ ◀ clean: × bottom, expand

STEP 2 $3nx - n = 3p + 4x$ ◀ collect the n terms

STEP 3+4 $n(3x - 1) = 3p + 4x \Rightarrow n = \frac{3p + 4x}{3x - 1}$ ◀ factorise n out, divide by the bracket

ANSWER $n = \frac{3p + 4x}{3x - 1}$

✓ $p = 1, n = 2: x = \frac{5}{2}, \frac{3+10}{6.5} = 2$

**WATCH OUT**

$$\times n = \frac{3p + n + 4x}{3x}$$

$$\checkmark n = \frac{3p + 4x}{3x - 1}$$

An answer with n on both sides is not finished — n must appear only on the left.

**EXAM TIP**

3 marks: M1 for clearing the fraction, M1 for collecting and factorising, A1. Write the factorised line.

**Check yourself**

calculator first · answers on page 17

4.1

Nov 2021 P1H

Make f the subject of

$$w = \frac{2f+3}{8-f}$$

Answer:

4.2

Jun 2022 P2H

Make c the subject of

$$g = \frac{c+3}{4+c} - 7$$

Answer:

4.3

Practice · Medium

Make m the subject of

$$g - 3m = am + 5$$

Answer:

V5**VARIANT 5** MOST COMMON IN THE EXAM

SEEN IN PAST PAPERS

7 / 16

PRACTICE QUESTIONS: 3

Root Round a Fraction

A $\sqrt{\quad}$ over a fraction with the subject top and bottom: square, then $V4$.

Came up in: Jan 2020 P2H Q16 · Jan 2021 P2H Q14(b) · Jun 2023 P2H Q17(b) · Jun 2023 P2HR Q17 · Jun 2024 P2H Q15(a) · Nov 2024 P2H Q12(b) · Nov 2025 P1H Q18

 Your notes

? What is it?

IN PLAIN WORDS

The exam's favourite rearrangement: a root over a whole fraction, and the subject in the top **and** the bottom. **Square both sides** (cube for $\sqrt[3]{\quad}$) — the root is gone. Now it is exactly **V4**: $\times$ the bottom, collect, factorise, divide.

IT LOOKS LIKE

$$p = \sqrt{\frac{y+w}{3y-t}}$$

$$y = \sqrt{\frac{x+1}{x-4}}$$

$$y = \sqrt[3]{\frac{6+5x}{x+4}}$$

1 SQUARE

$$p^2 = \frac{y+w}{3y-t}$$

2 $\times$ THE BOTTOM

$$p^2(3y-t) = y+w$$

3 COLLECT

$$3p^2y - y = w + p^2t$$

4 FACTORISE, $\div$

$$y = \frac{p^2t+w}{3p^2-1}$$

Four moves. Squaring first turns it into a V4 question.

How do I do it?

1 Square

Square both sides (cube for a cube root): $p \rightarrow p^2$.

2 $\times$ the bottom

$p^2(3y-t) = y+w$, then expand.

3 Collect

Subject terms on one side: $3p^2y - y = w + p^2t$.

4 Factorise, divide

$y(3p^2-1) = p^2t+w \rightarrow$ divide by $(3p^2-1)$.

SQUARE FIRST, THEN $V4$ The most common rearranging question.

V5 Root Round a Fraction

VARIANT 5 · SEE IT DONE, THEN TRY

 Your notes
 WHEN YOU SEE → YOU DO"Make y the subject of $p = \sqrt{\frac{y+w}{3y-t}}$ "

→ Square, × bottom, collect, factorise, divide

A cube root round the fraction

→ Cube both sides: y^3

(4 marks)

→ One mark per stage — write each line

 WORKED EXAMPLE **PAST PAPER**

November 2025 · Paper 1H · Q18 · 4 marks

Make y the subject of $p = \sqrt{\frac{y+w}{3y-t}}$

STEP 1 $p^2 = \frac{y+w}{3y-t}$ ◀ square both sides

STEP 2 $3p^2y - p^2t = y + w$ ◀ $\times(3y-t)$, expand

STEP 3 $3p^2y - y = w + p^2t$ ◀ collect the y terms

STEP 4 $y(3p^2 - 1) = p^2t + w \Rightarrow y = \frac{p^2t + w}{3p^2 - 1}$ ◀ factorise, divide

ANSWER $y = \frac{p^2t + w}{3p^2 - 1}$ ✓ $y = 1, w = 3, t = 1: p^2 = 2, \frac{2+3}{5} = 1$

 WATCH OUT

$$\times p^2(3y - t) = y + w \Rightarrow 3p^2y - t = y + w$$

$$\checkmark 3p^2y - p^2t = y + w$$

The p^2 multiplies **both** terms in the bracket.

 EXAM TIP

7 of the 16 past-paper parts. The answer can look different but equal:

$$\frac{p^2t+w}{3p^2-1} = \frac{-p^2t-w}{1-3p^2}$$

 Check yourself

calculator first · answers on page 17

5.1 Jan 2020 P2HMake x the subject of

$$y = \sqrt{\frac{x+1}{x-4}}$$

Answer:

5.2 Nov 2024 P2HMake c the subject of

$$f = \sqrt{\frac{a+bc}{c-d}}$$

Answer:

5.3 Jun 2023 P2HRMake x the subject of

$$y = \sqrt[3]{\frac{6+5x}{x+4}}$$

Answer:

V6

VARIANT 6

Squared Subject Twice t^2 on top and bottom: treat t^2 as one block, then root at the end.

SEEN IN PAST PAPERS

3 / 16

PRACTICE QUESTIONS: 2



Your notes

Came up in: Jan 2022 P2HR Q15 • Jun 2025 P2H Q16 • Jun 2025 P2HR Q17

? What is it?**In plain words**

The subject appears twice, but **always as t^2** (or t^3). Treat t^2 as **one block**: do $\sqrt{\quad}$ to get $t^2 = \dots$, then **root at the end**.

(If it were t^2 and t , you would complete the square — see page 3.)

IT LOOKS LIKE

$$c = \frac{t^2 + 3}{7 - 8t^2}$$

$$n^2 = \frac{4d + t^3}{t^3}$$

$$c = \frac{t^2 + 3}{7 - 8t^2}$$

↓ $\sqrt{\quad}$ WITH THE BLOCK

$$t^2 = \frac{7c - 3}{8c + 1}$$

↓ ROOT LAST

$$t = \sqrt{\frac{7c - 3}{8c + 1}}$$

How do I do it?**1** × the bottom

$$c(7 - 8t^2) = t^2 + 3, \text{ expand.}$$

2 Collect the blocks

$$\text{All } t^2 \text{ terms one side: } 7c - 3 = t^2 + 8ct^2.$$

3 Factorise, divide

$$t^2(1 + 8c) = 7c - 3 \rightarrow t^2 = \frac{7c-3}{8c+1}.$$

4 Root last

$\sqrt{\quad}$ the whole fraction ($\pm$ if allowed); $\sqrt[3]{\quad}$ for t^3 .

t^2 IS ONE BLOCK Root only after the block is alone.

V6 Squared Subject Twice

VARIANT 6 · SEE IT DONE, THEN TRY

 Your notes
 WHEN YOU SEE → YOU DO t^2 in the top and $8t^2$ in the bottom → Block t^2 : V4, then $\sqrt{\quad}$ t^3 twice: $n^2 = \frac{4d+t^3}{t^3}$ → Block t^3 , then $\sqrt[3]{\quad}$ "Given that $n > 0$, make n the subject" → Positive root only**WORKED EXAMPLE** **PAST PAPER**

June 2025 · Paper 2H · Q16 · 4 marks

Make t the subject of the formula $c = \frac{t^2 + 3}{7 - 8t^2}$ **STEP 1** $7c - 8ct^2 = t^2 + 3$ ◀ $\times(7 - 8t^2)$, expand**STEP 2** $7c - 3 = t^2 + 8ct^2$ ◀ collect the t^2 blocks**STEP 3** $t^2(1 + 8c) = 7c - 3 \Rightarrow t^2 = \frac{7c - 3}{8c + 1}$ ◀ factorise, divide**STEP 4** $t = \sqrt{\frac{7c - 3}{8c + 1}}$ ◀ root last

ANSWER $t = \sqrt{\frac{7c - 3}{8c + 1}}$

$\checkmark t = 1: c = -4, \sqrt{\frac{-31}{-31}} = 1$

**WATCH OUT**✗ rooting first: $t = \sqrt{\frac{\dots}{\dots}}$ term by term✓ root only when t^2 is alone

Square roots never split over + and -.

**EXAM TIP**Give $\pm$ unless the question says the subject is positive. Keep one fraction under one root.**Check yourself**

calculator first · answers on page 17

6.1

Jan 2022 P2HR

Make t the subject of

$$n^2 = \frac{4d + t^3}{t^3}$$

Answer:

6.2

Jun 2025 P2HR

Make k the subject of

$$p = \frac{8k^2 + 5}{7 - 3k^2}$$

Answer:

6.3

Practice · Hard

Given $n > 0$, make n the subject of $y = \frac{n^2 + d}{n^2}$

Answer:

CHECK YOURSELF

Answers



Every answer below was checked with sympy (check_answers.py). Got one wrong? Go back to that variant's Watch-out box first.

$$1.1 \quad a = \frac{d - g}{2c}$$

$$1.2 \quad a = \frac{M + bd}{c}$$

$$1.3 \quad t = \frac{v - u}{a}$$

$$2.1 \quad t = 3(y + 2a)$$

$$2.2 \quad F = \frac{9C + 160}{5}$$

$$2.3 \quad c = \frac{b}{a - 5}$$

$$3.1 \quad t = \sqrt[3]{c + 8v}$$

$$3.2 \quad t = \pm \sqrt{\frac{2s}{a}}$$

$$3.3 \quad k = 4m^2 - 1$$

$$4.1 \quad f = \frac{8w - 3}{w + 2}$$

$$4.2 \quad c = -\frac{4g + 25}{g + 6}$$

$$4.3 \quad m = \frac{g - 5}{a + 3}$$

$$5.1 \quad x = \frac{4y^2 + 1}{y^2 - 1}$$

$$5.2 \quad c = \frac{a + df^2}{f^2 - b}$$

$$5.3 \quad x = \frac{6 - 4y^3}{y^3 - 5}$$

$$6.1 \quad t = \sqrt[3]{\frac{4d}{n^2 - 1}}$$

$$6.2 \quad k = \pm \sqrt{\frac{7p - 5}{3p + 8}}$$

$$6.3 \quad n = \sqrt{\frac{d}{y - 1}}$$

THE WHOLE TOPIC ON ONE PAGE

Rearranging Formulas cheat sheet

**V1 Subject once 2/16**

$$p = at - d \Rightarrow t = \frac{p+d}{a}$$

Undo in reverse order.

V2 In a fraction 1/16

$$A + 5z = \frac{c}{y} \Rightarrow c = y(A + 5z)$$

Fraction alone, × the bottom.

V3 Powers & roots

$$p = \pm \sqrt{\frac{y+4}{3}}$$

Root round all → square first. Squared subject → $\pm\sqrt{\quad}$ last.**V4 Subject twice 3/16**

$$n(3x - 1) = 3p + 4x$$

× bottom → collect → factorise → divide.

V5 Root round a fraction 7/16

$$p^2 = \frac{y+w}{3y-t}$$

Square first, then V4.

V6 Squared twice 3/16

$$t^2(1 + 8c) = 7c - 3$$

 t^2 is one block; root last.**TOP 3 LOST MARKS**

1. $p^2(3y - t) = 3p^2y - t \times$
2. the subject still on both sides $\times$
3. rooting term by term $\times$

**QUADRATIC IN THE SUBJECT** x^2 and x both there? Complete the square first:

$$y = x^2 - 4x + 1 = (x - 2)^2 - 3 \Rightarrow x = 2 \pm \sqrt{y + 3}.$$

BOTTOM LINE Clean (root · fraction · bracket) → Isolate (collect · factorise · divide) → check with numbers.